TRIGONOMETRY NOTES

Lecture 1 Notes

TRIG001-01

Lecture 1: Similarity and Proportion

Algebra Review

Proportion: One ratio or one fraction equals another.

Example 1: $\frac{3}{4} = \frac{6}{8}$

$\frac{3}{4} \times \frac{6}{8}$ cross multiplication

$3 \cdot 8 = 4 \cdot 6$

$24 = 24$ equal ✓

TH

TRIG001-02

Lecture 1: Page 2

Example 2: $\frac{3}{5} \stackrel{?}{=} \frac{5}{8}$

$3 \cdot 8 = 5 \cdot 5$

$24 \neq 25$ not equal

Example 3: $\frac{x}{5} = \frac{3}{7}$

$\frac{7 \cdot x}{7} = \frac{15}{7}$

$x = \frac{15}{7}$

(Improper fractions are easier to deal with.)

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TRIG001-03

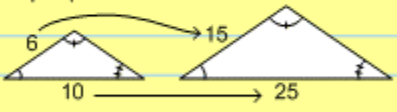
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TRIGONOMETRY

→ (measuring)

→ (figure having 3-sides)

Corresponding angles of similar triangles are congruent. However, the corresponding sides are not congruent, but proportional.



$\frac{6}{15} = \frac{10}{25}$ $\frac{\text{side of small triangle}}{\text{side of big triangle}}$

This is called an external ratio.

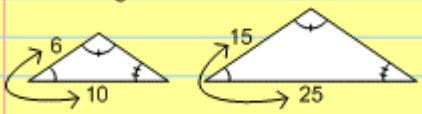
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TRIG001-04

Lecture 1: Page 4

An external ratio is a ratio of the length of a side of one triangle to the corresponding side of another triangle.

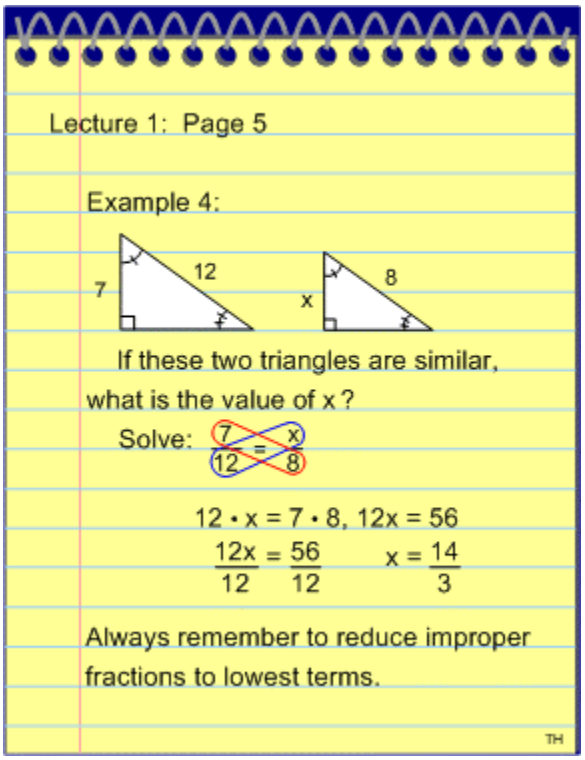
In trigonometry, however, we usually look at the ratio of sides within one triangle.



$\frac{6}{10} = \frac{15}{25}$

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TRIG001-05


Lecture 1: Page 5
Example 4:
7
12
x
8
If these two triangles are similar, what is the value of x ?
Solve: 7/12 = x/8
12 • x = 7 • 8, 12x = 56
12x/12 = 56/12
x = 14/3
Always remember to reduce improper fractions to lowest terms.
TH

TRIG002-01

Lecture 2: Right Triangles: 30-60-90

An equilateral triangle is a triangle having three sides of equal length. All three angles also have the same measure.

Recall that the sum of the angles of a triangle equals 180°.

In an equilateral triangle, all three angles have a measure of 60°.

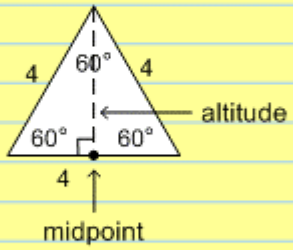
$$\frac{180^\circ}{3} = 60^\circ$$

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TRIG002-02

Lecture 2: Page 2

A line perpendicular to the base of a triangle is called the altitude.



The midpoint divides the base into two equal parts.

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TRIG002-03

Lecture 2: Page 3

Half of this equilateral triangle is a 30-60-90 right triangle.



How can we solve for x?

We can use the Pythagorean Theorem.

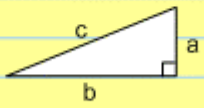
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TRIG002-04

Lecture 2: Page 4

Pythagorean Theorem

For any right triangle, there are two shorter sides (called legs) and one longest side (opposite the right angle) called the hypotenuse.



$$a^2 + b^2 = c^2$$

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